

Overview

1. Motivation / Goal to Achieve

Motivation: By 2030, the world will face a shortage of over 10 million healthcare workers — not because expertise doesn't exist, but because it is not intelligently coordinated. This is a planetary-scale systems failure and a moonshot opportunity for AI.

Rooted in a real-world challenge faced by the **Virtue Foundation**, this problem is not theoretical: skilled doctors remain disconnected from the hospitals and communities that urgently need them.

This challenge invites you to build an **agentic AI intelligence layer for healthcare** — a system that can reason, decide, and act to connect the right medical expertise with the right hospitals at the right moment.

You are not just building software. You are designing the coordination engine for global healthcare — turning data into action and multiplying human impact through AI.

Ambitious Goal: The goal of this challenge is nothing less than to **reduce the time it takes for patients to receive lifesaving treatment by 100×** using an agentic AI system.

You will build an **Intelligent Document Parsing (IDP) agent** that goes far beyond simple search. This agent will extract and verify medical facility capabilities from messy, unstructured data — and reason over it to understand where care truly exists and where it is missing.

Your system should be able to:

- Identify infrastructure gaps and medical deserts
Detect incomplete or suspicious claims about hospital capabilities
- Map where critical expertise is available — and where lives are currently at risk due to lack of access

If successful, your AI agent becomes part of a new healthcare intelligence layer: one that helps route patients and doctors faster, guide investment to the right places, and enable organizations like the Virtue Foundation to act with unprecedented speed and precision.

At planetary scale, even small improvements in coordination mean millions of patients treated sooner — and countless lives saved.

2. Core Features (MVP)

1. **Unstructured Feature Extraction:** Process free-form text fields (e.g., procedure, equipment, and capability columns) to identify specific medical data.
2. **Intelligent Synthesis:** Combine unstructured insights with structured facility schemas to provide a comprehensive view of regional capabilities.
3. **Planning system:** Think creatively how you could include a planning system which is easily accessible and could get adopted across experience levels and age groups

3. Stretch Goals (We know that you can do it!)

1. **Citations:** include row-level citations to indicate what data was used to support a claim by an agent.

Bonus: Provide citations at the *agentic-step level* (e.g., if your agent makes 5 sequential reasoning calls, demonstrate which specific data was used for each step).

Hint: Consider how experiment tracking tools can "trace" the inputs and outputs of internal agent loops to provide this level of transparency.

2. **Visualize with a Map:** create a map to demonstrate your conclusions visually. [Here](#) is some inspiration.
3. **Real-impact Bonus:** the Databricks team is collaborating with the VF team to ship an agent by June 7th. [Here are the questions being explored](#). Here are some real-world requirements that unlock impact and we invite you to tackle any of these!

4. Hints and Resources

- **Primary Tech Stack:** Focus on RAG and Agentic workflows, which are top-tier use cases for AI in the medical industry. Here's what the classic base state might be:
 - Agentic orchestrator: langgraph, llama index, crewAI
 - ML lifecycle: MLFlow
 - RAG: Databricks, FAISS, lanceDB
 - Text2SQL: genie
- **Environment:** The challenge is scoped for small amounts of high-impact medical data to ensure compatibility with the Databricks Free Edition.
- **Datasets:** Real-world facility reports and medical notes from a single country provided by the Virtue Foundation.
 - [Virtue Foundation Ghana Dataset](#)
 - [Schema Documentation](#)
 - [Prompts and pydantic models](#) used to create the data

5. Evaluation Criteria

- **Technical Accuracy (35%):** How reliably does the agent handle "Must Have" queries and detect anomalies in facility data?
- **IDP (Intelligent Document Parsing) Innovation (30%):** How well does the solution extract and synthesize information from unstructured "free-form" text?
- **Social Impact (25%):** Does the prototype effectively identify "medical deserts" to aid in resource allocation?
- **User Experience (10%):** Is the interface intuitive for non-technical NGO planners using natural language?

6. Why It Matters

Every data point you extract represents a patient who could receive care sooner. By automating understanding from medical notes — the most critical AI agent use case in healthcare — hackers are creating the intelligence layer that can transform scarcity into coordinated action and bring lifesaving expertise to the world's most underserved regions.

Data Legend

Schema Documentation: All Columns and Definitions

Organization Extraction

- **ngos** (list[str]): NGO names present in the text. An NGO is any non-profit organization that delivers tangible, on-the-ground healthcare services in low- or lower-middle-income settings.
- **facilities** (list[str]): Healthcare facility names present in the text. A publishable healthcare facility is any physical site that is currently operating and delivers in-person medical diagnosis or treatment to patients.
- **other_organizations** (list[str]): Named entities that don't meet facility or NGO classifications.

Base Organization Fields (Shared)

Contact Information

- **name** (str): Official name of the organization. Use complete, unabbreviated form with proper capitalization, excluding business suffixes like "Ltd", "LLC", "Inc".
- **phone_numbers** (list[str]): The organization's phone numbers in E164 format (e.g. '+233392022664').
- **officialPhone** (str): Official phone number associated with the organization in E164 format. Must be the organization's primary official contact number.
- **email** (str): The organization's primary email address.

Web Presence

- **websites** (list[str]): Websites associated with the organization.
- **officialWebsite** (str): Official website associated with the organization. Only include the domain name, not the full URL, and must correspond to the organization's official website.
- **facebookLink** (str): URL to the organization's Facebook page.
- **twitterLink** (str): URL to the organization's Twitter profile.
- **linkedinLink** (str): URL to the organization's LinkedIn page.
- **instagramLink** (str): Instagram account URL.
- **logo** (str): URL linking to the organization's logo image.

General

- **yearEstablished** (int): The year in which the organization was established.

- **acceptsVolunteers** (bool): Indicates whether the organization accepts clinical volunteers.

Address

- **address_line1** (str): Street address only (building number, street name). Do NOT include city, state, or country here.
- **address_line2** (str): Additional street address information (apartment, suite, building name).
- **address_line3** (str): Third line of street address if needed.
- **address_city** (str): City or town name of the organization. Parse from comma-separated location strings if needed.
- **address_stateOrRegion** (str): State, region, or province of the organization. Parse from comma-separated location strings if needed.
- **address_zipOrPostcode** (str): ZIP or postal code of the organization.
- **address_country** (str): Full country name of the organization. Always extract if country or country code information is present, using contextual clues from URL domain, phone numbers, or website content if not explicitly stated.
- **address_countryCode** (str): ISO alpha-2 country code of the organization. Derive from country name if needed - this field is REQUIRED when country is known.

Facility-Specific Fields

- **facilityTypeId** (enum): Type of facility. Levels: **hospital**, **pharmacy**, **doctor**, **clinic**, **dentist**.
- **operatorTypeId** (enum): Indicates if the facility is privately or publicly operated. Levels: **public**, **private**.
- **affiliationTypeIds** (list[enum]): Indicates facility affiliations (one or more). Levels: **faith-tradition**, **philanthropy-legacy**, **community**, **academic**, **government**.
- **description** (str): A brief paragraph describing the facility's services and/or history.
- **area** (int): Total floor area of the facility in square meters.
- **numberDoctors** (int): Total number of medical doctors working at the facility.
- **capacity** (int): Overall inpatient bed capacity of the facility.

NGO-Specific Fields

- **countries** (list[str]): Countries where the NGO operates (array of ISO alpha-2 codes).
- **missionStatement** (str): The NGO's formal mission statement.
- **missionStatementLink** (str): A URL to the NGO's published mission statement.
- **organizationDescription** (str): A neutral, factual description derived from the mission statement. Removes explicitly religious or subjective language.

Medical Specialties

- **specialties** (list[str]): The medical specialties associated with the organization. Must use exact case-sensitive matches from the specialty hierarchy (e.g., `internalMedicine`, `familyMedicine`, `pediatrics`, `cardiology`, `generalSurgery`, `emergencyMedicine`, `gynecologyAndObstetrics`, `orthopedicSurgery`, `dentistry`, `ophthalmology`). Choose the most specific appropriate specialty and only predict specialties clearly mentioned or strongly implied.

Facility Facts (Free-form)

- **procedure** (list[str]): Specific clinical services performed at the facility—medical/surgical interventions and diagnostic procedures and screenings (e.g., operations, endoscopy, imaging- or lab-based tests). Each fact should be a clear, declarative statement including specific quantities when available (e.g., "Offers hemodialysis treatment 3 times weekly").
- **equipment** (list[str]): Physical medical devices and infrastructure—imaging machines (MRI/CT/X-ray), surgical/OR technologies, monitors, laboratory analyzers, and critical utilities (e.g., piped oxygen/oxygen plants, backup power). Include specific models when available (e.g., "Has Siemens SOMATOM Force dual-source CT scanner").
- **capability** (list[str]): Medical capabilities defining what level and types of clinical care the facility can deliver—trauma/emergency care levels (e.g., "Level II trauma center"), specialized units (ICU/NICU/burn unit), clinical programs (stroke care, IVF), diagnostic capabilities, clinical accreditations (e.g., "Joint Commission accredited"), care setting (inpatient/outpatient), staffing levels, and patient capacity. Excludes addresses, contact info, business hours, and pricing.